RASHMI PANDEY

Software Engineer Candidate • Systems, Performance Optimization & Cloud Infrastructure

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EDUCATION

Amity University Jharkhand — Bachelor of Technology (B.Tech) in Computer Science & Engineering `2024 – 2028`

CGPA: 8.00 / 10.0 • Senior Secondary (Class XII): 88.2% • Secondary (Class X): 85.0% • Relevant Coursework: Operating Systems, Computer Architecture, Data Structures & Algorithms, Database Management Systems

TECHNICAL SKILLS

Programming Languages:	C, C++, C#, Java (Java 17/8, Multithreading), Python 3, SQL (T-SQL, MySQL), JavaScript (ES6+)
Systems & Performance:	Performance Analysis, Latency Optimization, Telemetry & Trace Evaluation, Memory Management, Concurrency, Debugging
Operating Systems & Core:	OS Internals, Process & Thread Scheduling, Resource Allocation, File Systems, I/O Subsystems, Linux/Unix, Windows
Cloud & Data Platforms:	Microsoft Azure (Azure SQL, Cloud Monitoring), AWS (EC2, CloudWatch, S3, IAM), Docker, Git, PM2, Maven
Databases & Architecture:	Azure SQL Database, MySQL (InnoDB B-Tree Indexing, ACID), MongoDB, Redis, RESTful Microservices, Jakarta EE

ENGINEERING & SYSTEMS RESEARCH EXPERIENCE

National Institute of Advanced Manufacturing Technology (NIAMT), Ranchi `Jun 2026 – Jul 2026`

Research Intern — Distributed Systems & Performance Optimization (Guided by Dr. Rahul Mishra)

- Architected **DVQ**, an intelligent workload routing and resource optimization system evaluated across 7 cloud regional topologies, analyzing live performance telemetry and traces to understand system behavior at scale.
- Formulated request scheduling via Lyapunov mathematical optimization, balancing strict sub-100ms latency SLAs against real-time carbon intensity and compute energy overhead.
- Investigated system bottlenecks using latency profiling and telemetry metrics; executed synthetic chaos testing under simulated regional grid outages, sustaining a verified **0.0% SLA breach rate**.
- Achieved **sub-100µs median decision latency** under Linux process supervision, validating system stability, zero memory leaks, and optimal CPU core utilization.

KEY ENGINEERING PROJECTS

EXAMORA — High-Throughput Academic Testing Engine (Java 17, Jakarta Servlets 6.0, MySQL, Tomcat 11) `github.com/rashmipandey870/EXAMORA`

- Engineered a native server-side testing engine utilizing the **Data Access Object (DAO)** architectural pattern over raw JDBC connection pooling, eliminating framework abstraction overhead for high-concurrency throughput.
- Analyzed SQL query execution plans to eliminate table scans; implemented composite B-Tree indexes on (exam_id, student_id) and row-level locks, guaranteeing zero transaction deadlock during peak load.
- Secured credentials using BCrypt hashing (salt rounds = 12); integrated Apache PDFBox pipeline to ingest and parse question bank documents with automated error recovery. (60+ structured Git commits).

Systems & Algorithms Lab — PriorityQueue Min-Heap Algorithmic Scheduler (Java, Data Structures) `rashmipandey870.github.io`

- Implemented an O(N log N) min-heap algorithmic scheduler evaluating task priority tradeoffs; built an interactive performance visualizer benchmarking memory page reads, cache efficiency, and execution cycles.

Sangathan — Algorithmic Tournament Scheduling Platform (Android Java, Node.js, Express, MongoDB) `github.com/rashmipandey870/Sangathan`

- Engineered an automated fixture platform generating collision-free cyclic round-robin and binary tree knockout brackets for 100+ athletes across 6 simultaneous sports with sub-150ms REST API response times.

CERTIFICATIONS & TECHNICAL ACHIEVEMENTS

• **Microsoft Certified: Azure Database Administrator Associate** (Credential ID: **647C0C2C5B35FA32**) — Certified by Microsoft in database performance analysis, query execution plan tuning, high-availability architectures, indexing, and resource optimization.

• **NPTEL Elite + Silver (Top 2% Nationwide): Python for Data Science** — **IIT Madras** (Score: **84%**). Applied statistical analysis, exploratory data analysis, and predictive modeling algorithms.

• **NPTEL Elite + Silver: The Joy of Computing using Python** — **IIT Ropar** (Score: **87%**).

• **100 Days of Code (Data Structures & Algorithms):** Solved **175+ algorithmic problems** across Trees, Graphs, Dynamic Programming, and Heaps. Maintained consistent open-source contribution with **1,200+ GitHub commits**.